

Sajib Acharjee Dip

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PHD RESEARCHER — LARGE LANGUAGE MODELS, POST-TRAINING & MULTIMODAL AI

PhD researcher in Computer Science at Virginia Tech working on large language models, post-training, multimodal learning, and agentic AI. I develop reinforcement-learning and verifier-based methods, tool-using agents, foundation models, and scalable Python/PyTorch pipelines for language, vision, biological, and drug-discovery problems.

WORK EXPERIENCE

Incoming Applied Scientist Intern (Amazon Music, USA)

Jan 2027 – Apr 2027

- Selected for an Applied Scientist internship focused on reinforcement learning, supervised learning, ranking, sequential modeling, and large-scale recommendation systems.
- Planned work focuses on learning user representations and optimizing recommendation decisions from large-scale behavioral signals.

AI Research Intern (Genentech, USA)

May 2026 – Aug 2026

- Developed agentic AI systems integrating LLMs with external tools, retrieval, scientific databases, and automated research workflows.
- Built MCP-based tool integrations and reusable infrastructure for scientific data retrieval, reasoning, orchestration, and model evaluation.
- Developed scalable multimodal ML pipelines combining pathology images, molecular data, and structured biological evidence.
- Collaborated with scientists and engineers to translate drug-discovery questions into computational experiments and reusable AI systems.

Research Assistant (Virginia Tech, USA)

May 2024 – Present

- Develop research across LLM reasoning, post-training, reinforcement learning, multimodal learning, foundation models, and agentic AI.
- Investigated reward hacking and reward-quality divergence in vision-language reinforcement learning, including cases where optimization improves learned rewards without improving true model quality.
- Developed post-training methods using policy optimization, reward modeling, verifier-guided learning, selective verification, and inference-time reasoning.
- Built multimodal models and evaluation frameworks spanning text, images, graphs, biological sequences, and high-dimensional scientific data.
- Developed multi-agent and retrieval-augmented systems combining LLMs, external tools, structured knowledge, and iterative verification.
- Built scalable Python/PyTorch pipelines for model training, inference, evaluation, retrieval, and large-scale experimentation on Linux/HPC systems.

Summer Research Intern (Labcorp, USA)

May 2025 – Aug 2025

- Developed machine-learning pipelines for large-scale genomic, clinical, and biomarker datasets.
- Built reproducible Python workflows for data processing, statistical modeling, feature engineering, and downstream scientific analysis.
- Collaborated with scientists to translate biological questions into measurable computational objectives and experiments.

Machine Learning Engineer (IQVIA, USA)

May 2022 – Jul 2023

- Designed and implemented machine-learning solutions for large-scale real-world healthcare datasets.
- Built automated data-processing and ML pipelines using Python, SQL, and AWS.
- Developed predictive modeling, forecasting, and statistical-learning workflows for production-oriented applications.
- Collaborated with engineering and domain teams to build scalable, reusable ML and data workflows.

Research Intern @ Xu Lab (Carnegie Mellon University, USA)

Apr 2021 – Jul 2023

- Developed deep representation-learning methods for large-scale 2D and 3D imaging datasets.
- Applied object detection, segmentation, structured prediction, representation learning to complex visual data.
- Built scalable deep-learning training and evaluation pipelines using PyTorch and HPC infrastructure.

Research Intern (Harvard University)

May 2022 – Jan 2023

- Developed deep-learning and representation-learning methods for high-dimensional biomedical and visual data.
- Applied semi-supervised learning and feature representation methods to image-analysis problems.
- Trained and evaluated deep-learning models using large-scale HPC infrastructure.

EDUCATION

- **Virginia Tech** Blacksburg, USA
Ph.D. Student, Computer Science [Supervisor Dr. Liqing Zhang] Aug 2023 - May 2027(Expected)
- **Virginia Tech** Blacksburg, USA
Masters in Computer Science [Supervisor Dr. Liqing Zhang]; CGPA: 4.00/4.00 Aug 2023 - April 2025
Courses Taken: Natural Language Processing, Multimodal Computer Vision, Advanced Computational Biology, Software Engineering, Statistics in Research
Research Topic: Generative AI for Disease Diagnosis and Drug Discovery using Multimodal Biomedical Data.
(UnCOT-AD, PathoLM, and MoXGATE were developed as part of my Master's and PhD research at Virginia Tech)
- **Bangladesh University of Engineering and Technology** Dhaka, Bangladesh
Bachelor of Computer Science and Engineering; CGPA: 3.84/4.00 (Marks: 86.56%) Feb 2017 - May 2022
Courses Taken: Machine Learning, Bioinformatics, Artificial Intelligence, Mathematical Analysis for CS, Data Structure and Algorithms, Computer Security, Linear Algebra, Graph Theory, Operating System

SELECTED PUBLICATIONS

4.1 Agentic AI, Reasoning & Representation Learning

Looking or Talking? How Vision-Language Models Ground Chain-of-Thought Reasoning, , SA Dip, L Zhang, X Wang, T Vu, (Under Review)

Catching Reward Hacking in the Act: Reward-Quality Divergence for Vision-Language Reinforcement Learning, SA Dip, L Zhang, T Vu, (Under Review)

Trust-Weighted Policy Optimization from Stress-Tested Rubric Rewards, SA Dip, T Vu, (Under Review)

Think Again or Think Longer? Selective Verification for Budget-Aware Reasoning , SA Dip, D Zhou, L Zhang — (Under Review) [\[Preprint\]](#) [\[Hugging Face\]](#)

ProtFunAgent: Agentic LLM Cascades for Low-Resource Protein Function Gap-Filling via Homology RAG and Ontology-Constrained Decoding , SA Dip, J. Choy, L Zhang — [Neurips 2025 ER](#) (Spotlight) [\[Paper\]](#)

NicheAgent: Constrained Language-Guided Refinement for Zero-Shot Spatial Annotation , SA Dip, L Zhang — [ICLR MLGenX 2026](#) [\[Paper\]](#)

BenSync: Benchmarking Conversational Sycophancy and Human Alignment in LLMs for Bengali Contexts, K Noshin, SA Dip, RD Prangan, FH Tamim, SI Ahmed, L Zhang, S Sultana — (Under Review) [\[Preprint\]](#) [\[HF\]](#)

PlantMarkerBench: A Multi-Species Benchmark for Evidence-Grounded Plant Marker Reasoning, SA Dip, S Li, L Zhang — (Under Review) [\[Preprint\]](#) [\[Huggingface\]](#)

LLM4Cell: A Survey of Large Language and Agentic Models for Single-Cell Biology, SA Dip, A Zafor, BK Paul, UA Shuvo, MI Emon, X Wang, L Zhang — [ACL Main 2026](#) [\[Paper\]](#)

Large language model agents for biological intelligence across genomics, proteomics, spatial biology, and biomedicine, SA Dip, D Mallick, UA Shuvo, SB Soumma, F Rafsani, BK Paul, NA Moumi, S Ahmed, L Zhang — [Briefings in Bioinformatics](#) [\[Paper\]](#)

4.2 Generative and Foundation Models

Patch-Level Tissue Context Improves Learning from Frozen Pathology Foundation Model Embeddings, [SA Dip](#), L Zhang, (ACM-BCB 2026) [\[Paper\]](#)

Predicting Unseen Gene Perturbation Response Using Graph Neural Networks with Biological Priors, [SA Dip](#), L Zhang — (ACM-BCB 2026) [\[Paper\]](#)

CFM-GP : Unified Conditional Flow Matching to Learn Gene Perturbation Across Cell Types, AR Abir*, [SA Dip*](#), L Zhang — [\[Paper\]](#) (Nucleic Acids Research G&B) **Equal contrib.*

PathoLM: Identifying Pathogenicity of the DNA Sequence Through The Genome Foundation Model, [SA Dip](#), ..., Xuan Wang, L Zhang — [MLCB 2024](#) (Spotlight) [ICML 2024 - AI4Science](#) [\[Paper\]](#) [\[Code\]](#)

4.3 Multimodal Data Integration

PAINT: Paying Attention to INformed Tokens to Mitigate Hallucination in Large Vision-Language Model, KHI Arif, [SA Dip](#), K Hussain, L Zhang, C Thomas, [CVPR 2025 - TMM-OpenWorld](#) [\[Paper\]](#) [\[Code\]](#)

Mixture-of-Experts Guided Multi-Omic Integration for Gastrointestinal Cancer Subtype Prediction, [SA Dip](#), UA Shuvo, D Mallick, AR Abrar, L Zhang — [Neurips 2025 GenAI4Health](#) [\[Paper\]](#)

UnCOT-AD: Unpaired Cross-Omics Translation Enables Multi-Omics Integration for Alzheimer's Disease Prediction , AR Abir, [SA Dip](#), L Zhang, [Briefings In Bioinformatics](#) [\[Paper\]](#)

TECHNICAL SKILLS

Programming: Python, C/C++, SQL, Bash

Large Language Models: Transformers, LLM Architecture, Foundation Models, Instruction Tuning, SFT, QLoRA, Structured Generation, Retrieval-Augmented Generation

Post-Training & Reinforcement Learning: Reward Modeling, Policy Optimization, Verifier-Based Learning, RLHF, Selective Verification, Inference-Time Reasoning

Multimodal Learning: Vision-Language Models, Graph Neural Networks, Representation Learning, Multimodal Fusion, Biological Sequence Modeling

Agentic AI: Tool-Using Agents, ReAct, MCP, Multi-Agent Systems, Workflow Orchestration, Evaluation Harnesses

ML Systems: PyTorch, Hugging Face, Scikit-learn, Model Training & Inference, Large-Scale Experimentation, Data Pipelines, Linux/Unix, Docker, AWS, SLURM, HPC

LEADERSHIP AND COMMUNICATIONS

Conference Reviewer: AAAI 2027, Neurips 2026, ICML 2026, KDD 2026, ACM-BCB 2026, ACL Industry Track 2026, BHI 2025, ICLR Main 2025, ICLR AI4NA 2025, ICLR QUESTION 2025, Neurips GenAI4Health 2025, ISMB 2024 (Sub-Reviewer)

Journal Reviewer: Bioinformatics (Oxford, 2026), Omega (ACS, 2026), Genomics (Elsevier, 2025), GigaScience (Oxford, 2025), Computational Biology and Bioinformatics (SciencePG, 2025)

Program Committee: ICML SPIGM 2026, ISMB MLCSB 2026, Chen Institute Symposium for AI Accelerated Science 2025

Student Volunteer: GPSS Senator 2025, Committee Member of Undergrad Student Policy 2025, Student Volunteer at AAAI Fall Symposia 2024, Virginia Tech CS Graduate Recruitment (2024, 2025)

Presentation: ACL 2026, ICLR 2026, ICCABS 2026, NeurIPS 2025, Commonwealth of Virginia Cancer Research Conference (CVCRC) 2025, AAAI 2025, CCI Symposium 2025, MLCB 2024, AAAI FSS 2024